

Main Text — Tables and Figures

Racial and Ethnic Differences in Breast Cancer Incidence in the United States: A Systematic Review with Quantitative Synthesis

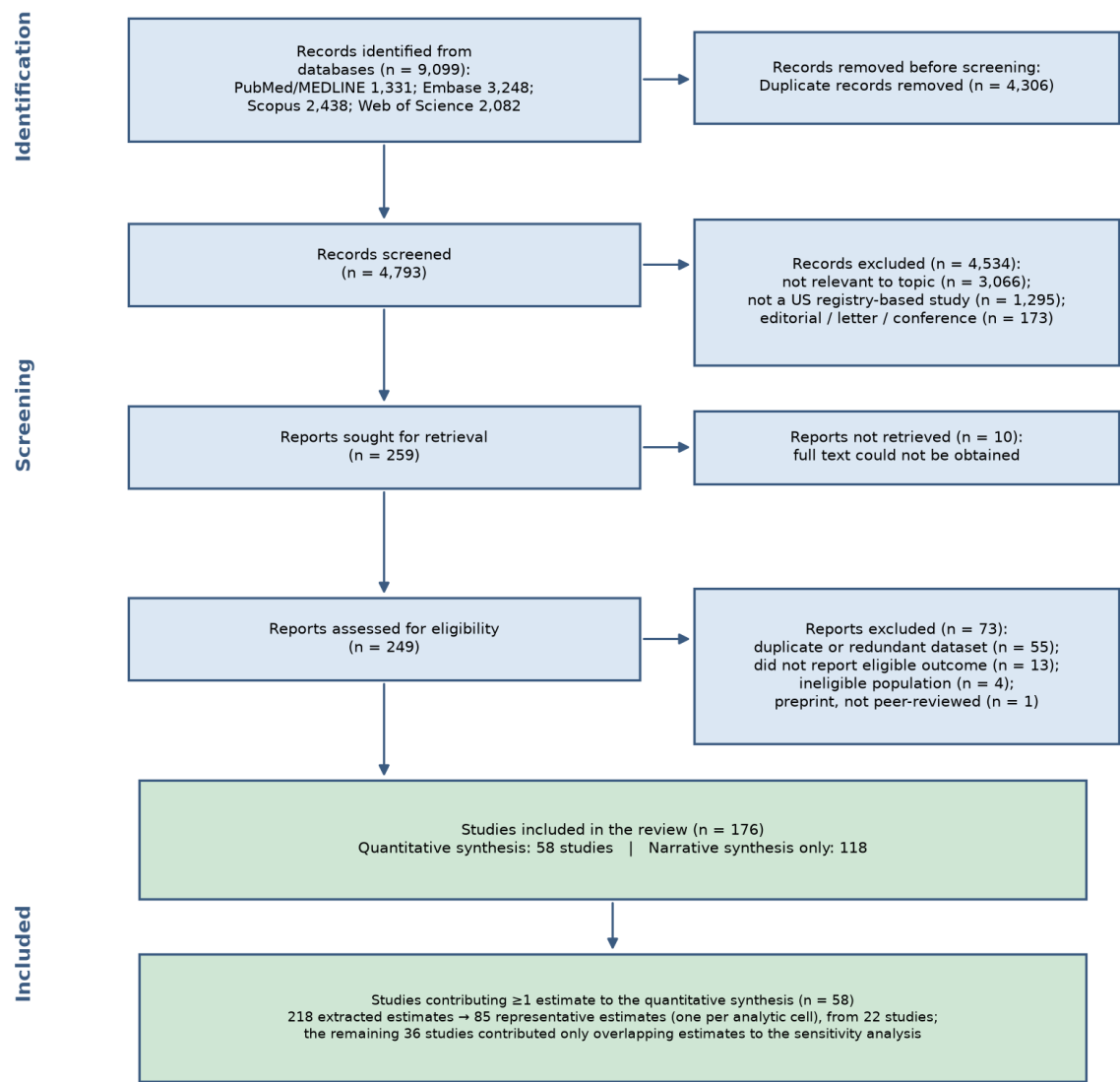
Table 1. Incidence rate ratios of invasive breast cancer among U.S. racial and ethnic groups relative to non-Hispanic White women, by analytic dimension

Group	Effect	Estimate [95% CI]	Representative study	Registry family	RoB
Overall invasive breast cancer					
Hispanic	IRR	0.718 [0.707, 0.731] ‡	Ellington 2022 (2018)	USCS(NPCR+SEER ~99%)	Low
Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	IRR	0.769 [0.752, 0.788] ‡	Ellington 2022 (2018)	USCS(NPCR+SEER ~99%)	Low
American Indian and Alaska Native (AI/AN)	IRR	0.87 (point est.)	Melkonian 2019 (2010-2015)	IHS-PRCDA	Moderate
non-Hispanic Black (NHB)	IRR	0.933 [0.920, 0.946] ‡	Ellington 2022 (2018)	USCS(NPCR+SEER ~99%)	Low
Asian American subgroups					
Hmong	IRR	0.164 [0.120, 0.224] ‡	Mills 2005 (1988-2000)	California-CCR	Low
Cambodian	IRR	0.264 [0.204, 0.341] ‡	Kem 2007 (1998-2002)	California-CCR	Moderate
Laotian/Kampuchean	IRR	0.388 [0.350, 0.430] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Vietnamese	IRR	0.586 [0.566, 0.608] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Korean	IRR	0.677 [0.654, 0.701] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Chinese	IRR	0.760 [0.745, 0.775] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Filipina	IRR	0.857 [0.839, 0.874] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Asian Indian/Pakistani	IRR	0.958 [0.937, 0.979] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Japanese	IRR	1.045 [1.009, 1.083] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Asian American (aggregate)	IRR	0.792 [0.784, 0.800] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Native Hawaiian and Pacific Islander (NHPI) subgroups					
Guamanian/Chamorro/Samoan	IRR	0.723 [0.641, 0.817] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Native Hawaiian	IRR	1.270 [1.213, 1.330] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Native Hawaiian and Pacific Islander (NHPI)	IRR	1.212 [1.168, 1.258] ‡	Gomez 2026 (2018-2022)	SEER-national	Moderate
Hispanic/Latina by country of origin					
Mexican	IRR	0.512 [0.382, 0.686] ‡	Pinheiro 2009 (1999-2001)	State: Florida	Low
New Latino	IRR	0.697 [0.629, 0.771] ‡	Pinheiro 2009 (1999-2001)	State: Florida	Low
Cuban	IRR	0.769 [0.688, 0.860] ‡	Pinheiro 2009 (1999-2001)	State: Florida	Low
Puerto Rican	IRR	0.833 [0.738, 0.939] ‡	Pinheiro 2009 (1999-2001)	State: Florida	Low
American Indian and Alaska Native (AI/AN) by region					
AI/AN (Navajo)	IRR	0.49 [0.44, 0.55]	Yazzie 2025 (2014-2018)	IHS-PRCDA	Low
AI/AN (Southwest)	IRR	0.57 (point est.)	Melkonian 2019 (2010-2015)	IHS-PRCDA	Moderate
AI/AN (East)	IRR	0.61 (point est.)	Melkonian 2019 (2010-2015)	IHS-PRCDA	Moderate
AI/AN (Pacific Coast)	IRR	0.93 (point est.)	Melkonian 2019 (2010-2015)	IHS-PRCDA	Moderate
AI/AN (Northern Plains)	IRR	1.05 (point est.)	Melkonian 2019 (2010-2015)	IHS-PRCDA	Moderate
Alaska Native	IRR	1.09 [0.99, 1.21] †	Nash 2019 (2009-2014)	Alaska Native Tumor Registry (IHS-linked)	Low
AI/AN (Southern Plains)	IRR	1.33 [1.26, 1.41]	Melkonian 2021 (2012-2016)	IHS-PRCDA	Low
Middle Eastern					
Middle Eastern	IRR	0.86 [0.84, 0.88]	Nasseri 2009 (1988-2004)	California-CCR	Moderate

Group	Effect	Estimate [95% CI]	Representative study	Registry family	RoB
Triple-negative breast cancer					
Chinese	IRR	0.53 [0.31, 0.93] † ‡	Loo 2019 (2010-2013)	Hawaii Tumor Registry (SEER)	Low
Asian American (aggregate)	IRR	0.61 [0.58, 0.65]	Gomez 2017 (2009-2013)	California-CCR	Low
Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	IRR	0.700 [0.680, 0.720]	Sung 2023 (2015-2019)	USCS(NPCR+SEER ~99%)	Low
Filipina	IRR	0.84 [0.60, 1.18] † ‡	Loo 2019 (2010-2013)	Hawaii Tumor Registry (SEER)	Low
American Indian and Alaska Native (AI/AN)	IRR	0.860 [0.800, 0.930]	Sung 2023 (2015-2019)	USCS(NPCR+SEER ~99%)	Low
Hispanic	IRR	0.860 [0.840, 0.880]	Sung 2023 (2015-2019)	USCS(NPCR+SEER ~99%)	Low
Native Hawaiian	IRR	0.86 [0.60, 1.22] † ‡	Loo 2019 (2010-2013)	Hawaii Tumor Registry (SEER)	Low
Japanese	IRR	1.08 [0.79, 1.46] † ‡	Loo 2019 (2010-2013)	Hawaii Tumor Registry (SEER)	Low
non-Hispanic Black (NHB)	IRR	1.950 [1.930, 1.980]	Sung 2023 (2015-2019)	USCS(NPCR+SEER ~99%)	Low

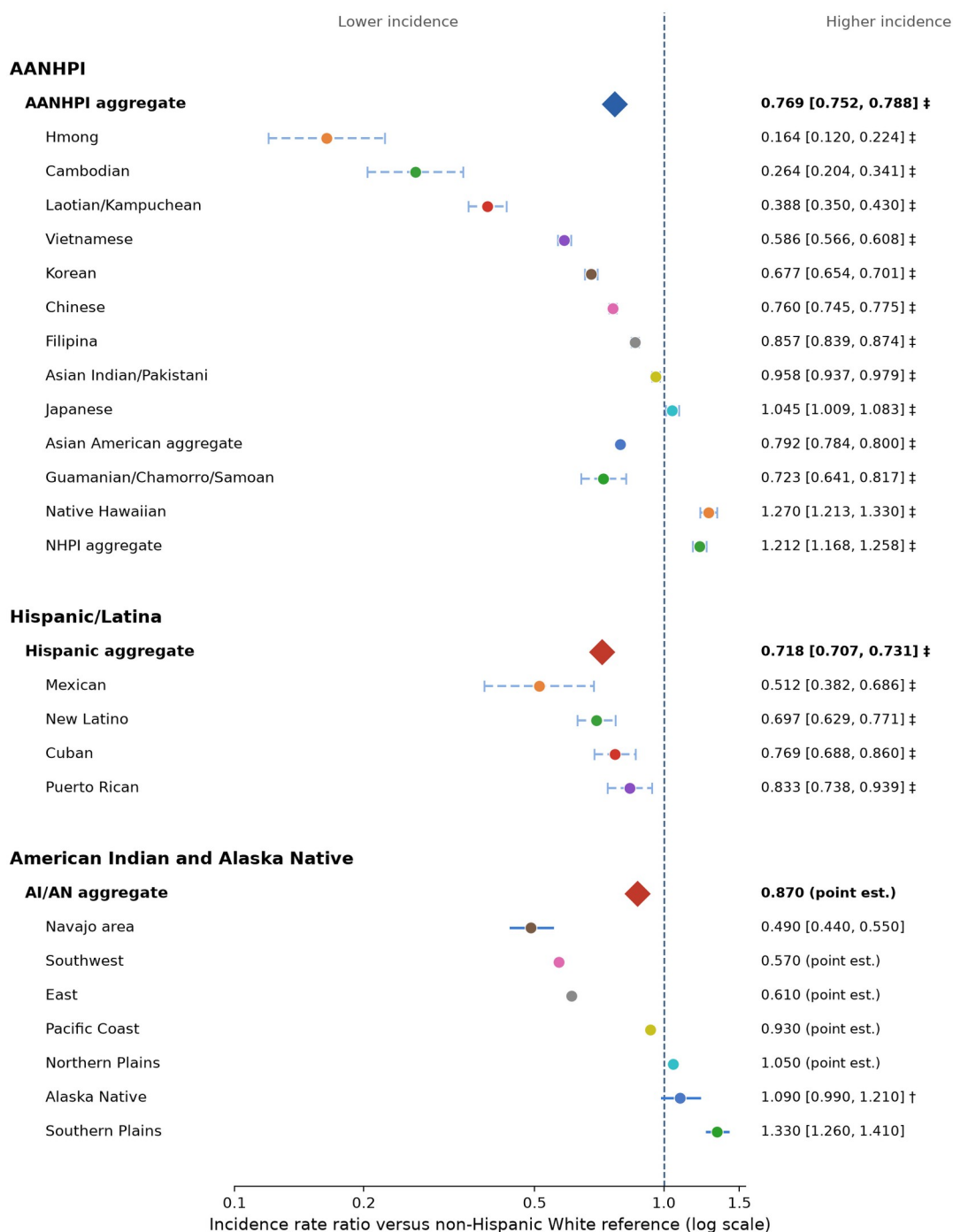
Note. Each cell shows one representative population-based estimate — the most recent, broadest-coverage registry estimate per analytic cell — not a pooled estimate; the selection rule and its robustness are given in the Methods and Supplementary Table 6. The effect measure is the incidence rate ratio (IRR) unless marked as a standardized incidence ratio (SIR); comparisons are versus non-Hispanic White (NHW) women. † the study's reference was an unstratified White group (not stratified by Hispanic origin), examined in the NHW-comparator sensitivity analysis (Supplementary Table 6c). ‡ the 95% confidence interval was computed by the reviewers from published rates rather than reported in the source. NHB, non-Hispanic Black; AANHPI, Asian American, Native Hawaiian, and Pacific Islander (NHPI, the Pacific Islander subset, shown separately); RoB, risk-of-bias rating of the representative study. Full per-estimate detail is in the Supplementary Materials.

Figure 1. PRISMA 2020 flow diagram



PRISMA 2020 flow diagram. Identification via databases only; no registers, citation searching, or grey-literature sources were used. Single-reviewer screening with AI assistance. Search date 7 August 2026.

Figure 2. Aggregate-to-disaggregated heterogeneity in breast cancer incidence (representative IRR vs non-Hispanic White, 95% CI): the Asian American, Native Hawaiian and Pacific Islander (AANHPI), Hispanic/Latina, and American Indian and Alaska Native (AI/AN) aggregates, each shown with its subgroups



Note. Each point is the representative estimate for one analytic cell, drawn from a separate study; an aggregate and its subgroups differ in registry, region, diagnosis period, and standard population. Diamonds mark aggregate groups, circles subgroups. A solid bar is a 95% CI reported in the source; a dashed bar marked ‡ is one computed for this review; a point without a bar had no interval in its source. ‡ reference was an unstratified White group, not NHW. Values match Table 1.

Figure 3. Incidence rate ratio by racial or ethnic group and analytic dimension (representative estimate vs non-Hispanic White; 1.0 = the NHW rate)

